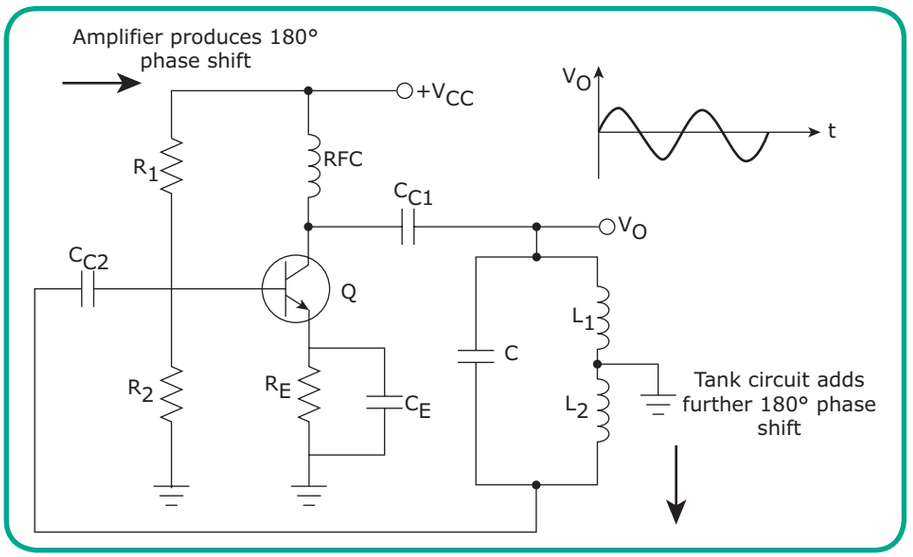


Figure 7.6 Hartley Oscillator

Working Principle: When the circuit is turned 'ON', the capacitor 'C' is charged. It is fully charged, starts discharging through the coils L_1 and L_2, which starts the oscillations. The output voltage of the amplifier appears across L_1 and the feedback voltage appears across L_2. The voltage across L_2 is 180⁰ out-off phase with the output voltage. It is a feedback signal and thus the total phase shift is 180⁰ + 180⁰ = 360⁰, since the output waveform provides 180⁰ phase shift and the feedback provides another 180⁰ phase shift. In this way, feedback is properly phased to produce continuous Undamped oscillation.

The oscillator frequency is calculated by the values of L_1, L_2 and C and is given by,

$$F = \left(\frac{1}{2\pi\sqrt{LC}}\right), \text{ where } L=L_1+L_2$$

Advantages

1. The frequency adjusted by single variable capacitor.
2. The output amplitude remains constant over the frequency range.

7.5.2. RC Phase Shift Oscillator

An oscillator which have resistive and capacitive element with a phase shift circuit is called RC phase shift oscillator. Previously, we have studied about LC circuits. They have two general drawbacks. Firstly, they suffer from frequency instability and poor waveform. Secondly, they cannot be used for low frequencies because they become too bulky and expensive.

History: Transistor phase shift oscillator was first patented by Robert W. Blanchard in the year 1958.

Construction: Figure 7.7 shows the circuit of phase shift oscillator. It consists of transistor amplifier and 'RC' phase shift network. The phase shift network has three sections R_1C_1, R_2C_2 and R_3C_3. At some particular frequency f_0, the phase shift in each 'RC' section is 60⁰ so that the total phase-shift produced by the three RC network is 180⁰.

The oscillator frequency

$$f_o = \frac{1}{2\pi RC\sqrt{6}},$$

where $R_1=R_2=R_3=R$ and $C_1=C_2=C_3=C$.

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